

Name: _____

Unit 2.1 - Counting and sequences

6Nc.01 Count on and count back in steps of constant size , including fractions and decimals, and extend beyond zero to include negative numbers.

11) Write the next term in the following sequences. (3 marks)

a) 0.3, - 0.2, - 0.7, - 1.2, _____

b) _____

c) 2.4, 2.45, 2.5, 2.55, _____

12) Mr. David counts on in steps of 0.002. He starts on 5.698 . What are the next two numbers in his count? (2 marks)

6Nc.03 Use the relationship between repeated addition of a constant and multiplication to find and use a position to term rule.

13) Mr. Conor recorded some information about a sequence in the table below.

Position	1	2	3	4	5
Term	6	12		24	

a) Complete the table (2 marks)

b) What is the term to term rule? _____ (1 mark)

c) What is the position to term rule? _____ (1 mark)

d) What is the 30th term in the sequence? _____ (1 mark)

Unit 2.2 - Special numbers

6Nc.04 - Use knowledge of square numbers to generate terms in a sequence given its position.

14) Complete the table. (4 marks)

a) What is the sum of the fourth square number and the second square number?	
b) What is the product of the second square number and the third square number?	
c) What is the difference between the ninth square number and the sixth square number?	
d) What is the seventh square number divided by the tenth square number.	

6Ni.08 - Use knowledge of multiplication and square numbers to recognise cube numbers (from 1 - 125).

15) Circle the following that are the same as 4^3 . (3 marks)

$4^2 \times 4$

$4 \times 4 \times 4 \times 4$

$4 \times 4 \times 4$

$4 + 4 + 4$

64

16) A number is cubed and then 5 is added. The answer is 13. What is the original number? (1 mark)

Unit 2.3 - Common factors and multiples.

6Ni.06 - Understand the common multiples and factors.

17) Mr. Matthew says, "I am thinking of a number that is a multiple of 6. My number is also a multiple of 7." Write the smallest three numbers that he could be thinking of. (3 marks).

18) Find **all** the common factors of 24 and 8. (1 mark)
